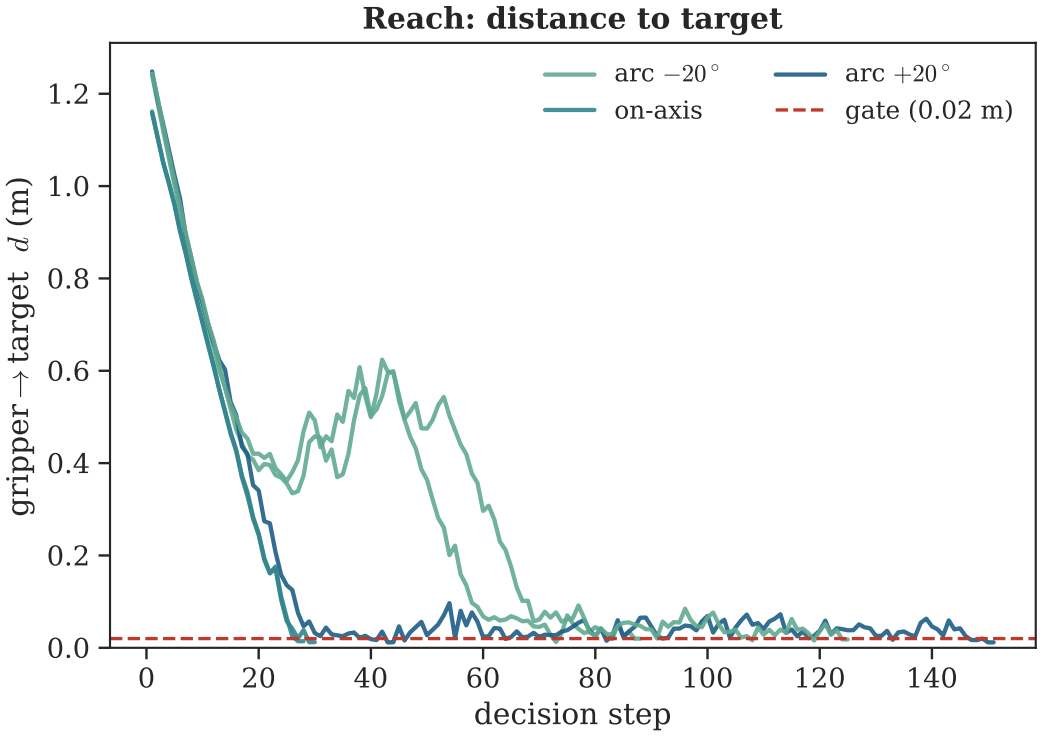
# s9_tarc — Figure Notes for the Defense

s9 direct-velocity whole-body (14/8), T-arc curriculum. Gate 0.02 m / 1.15°; 100% reached; mean min $d$ 13.2 mm, mean min $\varphi$ 0.39°, mean 83 decisions (8.3 s) to the gate. Checkpoint ppo_scoutsan_s9_tarc.zip, 2026-09-25.

## Figure 1 — Reach: distance to target



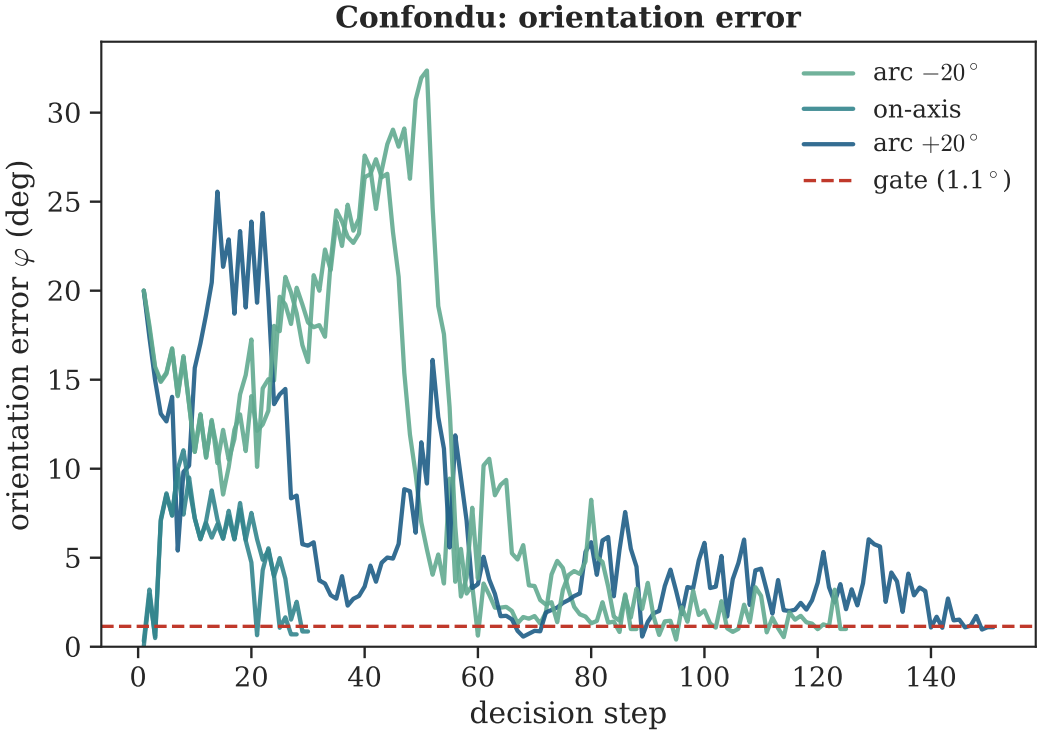

**What it shows.** Gripper→target distance $d$ vs decision step, one curve per pass, coloured by WHICH of the three curriculum targets it is (arc −20° / on-axis / arc +20°); red dashed = the 0.02 m position gate.

**How to read it.** Every pass converges under the gate (reached rate 100%). The curve stops at the gate because the agent ends the episode there.

**Say this.** "All 5 passes reach the 0.02 m gate, mean minimum distance 13.2 mm."

**If asked.** *Why do the curves end at different lengths?* — each curve ends when its target is reached; the on-axis target is far quicker than the ±20° arc ones.

## Figure 2 — Confondu: orientation error



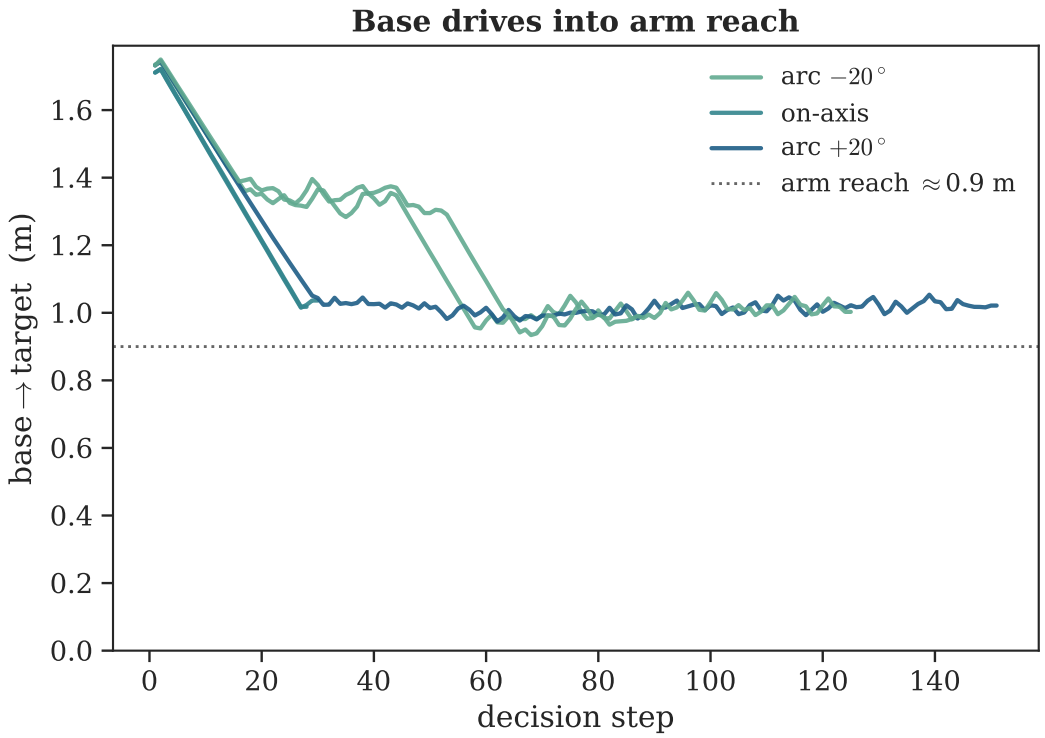

**What it shows.** Orientation error $\varphi$ (deg) vs step; gate = 1.15° (0.02 rad). $\varphi = 2 \arccos |\langle q_G, q_T \rangle|$, the geodesic angle between the gripper and target frames.

**How to read it.** $\varphi$ starts near 90° (the gripper faces its home pose, the target faces away from the robot) and falls under the gate together with the distance — that simultaneity is what *confondu* means.

**Say this.** "Orientation converges to a mean minimum of 0.39°, under the 1.15° gate."

**If asked.** *Why is position alone not enough?* — the task is frame coincidence: the gripper must arrive with the right orientation, not just at the right point.

## Figure 3 — Base drives into arm reach



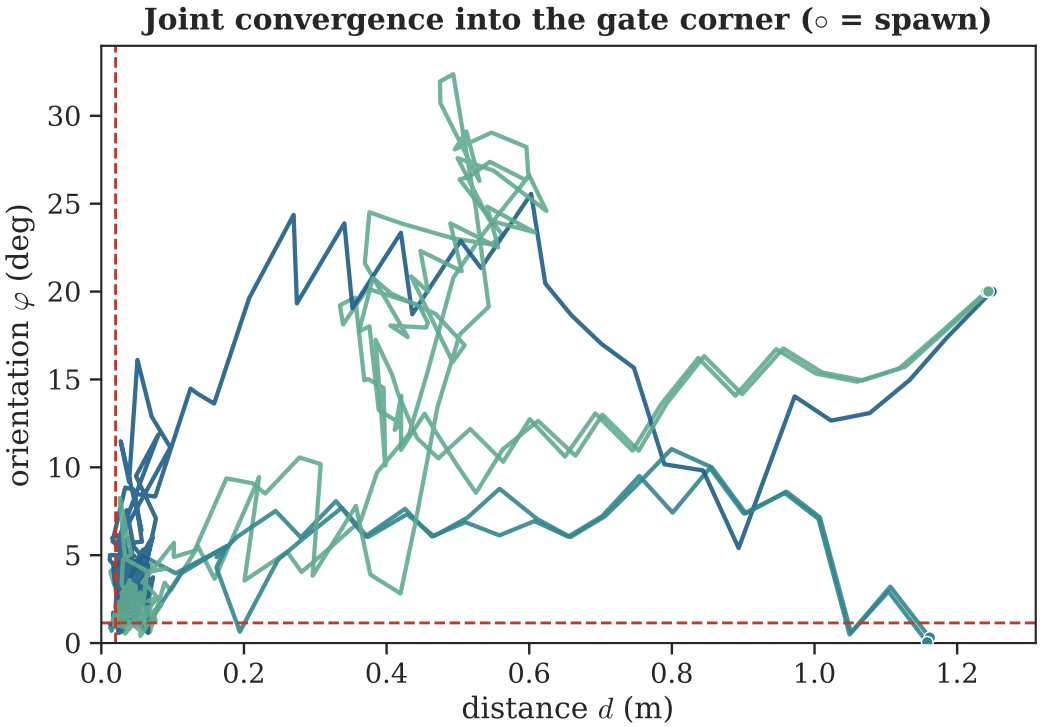

**What it shows.** base_link→target distance vs step — the mobile base's contribution, separated from the arm's. The dotted line marks the ≈0.9 m arm reach.

**How to read it.** The base closes from about 2 m to inside arm reach and then holds while the arm finishes; this is the whole-body split the single policy learned on its own (no hand-coded hand-off).

**Say this.** "One policy drives the base AND the arm: the base brings the target inside the arm's reach, then the arm performs the alignment."

**If asked.** *Is the split hand-coded?* — no: the 8 actions are 6 joint deltas plus base $v, \omega$; the division of labour is learned.

## Figure 4 — Joint convergence into the gate corner



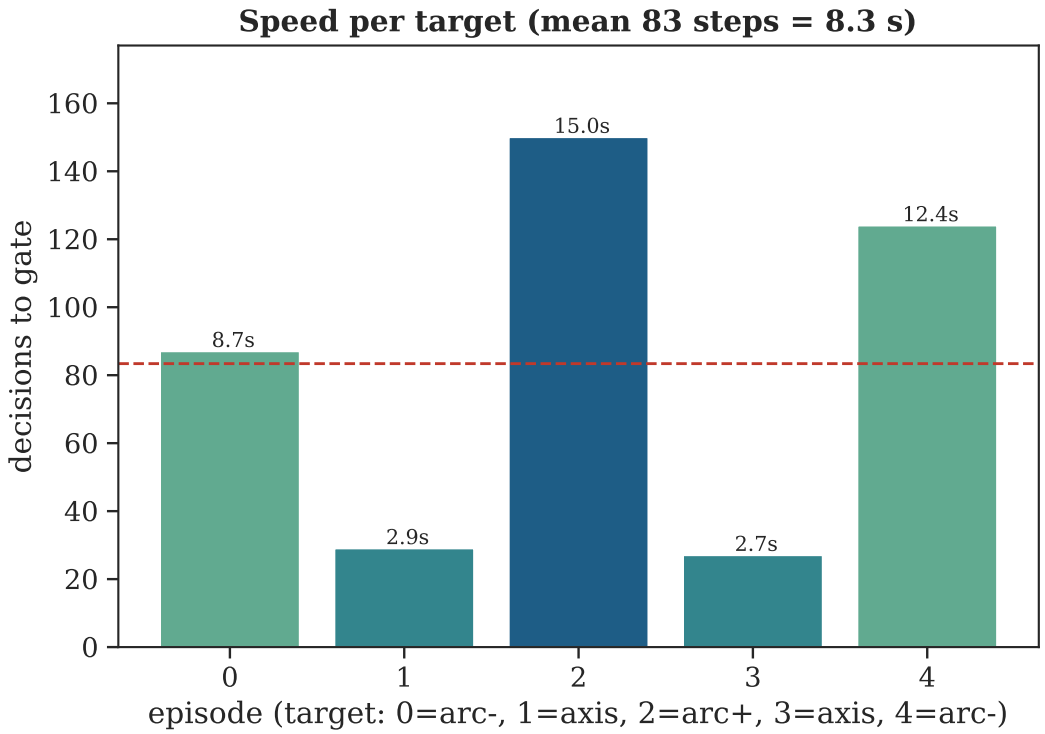

**What it shows.** $\varphi$ plotted against $d$ (a phase portrait); the open circle is the spawn, the red lines are the two gate thresholds. The bottom-left corner is *confondu*.

**How to read it.** The trajectories sweep down-left into the corner instead of hugging one axis — position and orientation are reduced together, not one after the other.

**Say this.** "Both errors enter the gate corner together — the policy does not trade position for orientation."

**If asked.** *What would a bad policy look like?* — a curve that runs down the $d$ axis first and only then drops in $\varphi$ (reach, then fix the wrist), which wastes time and often stalls.

## Figure 5 — Speed per target



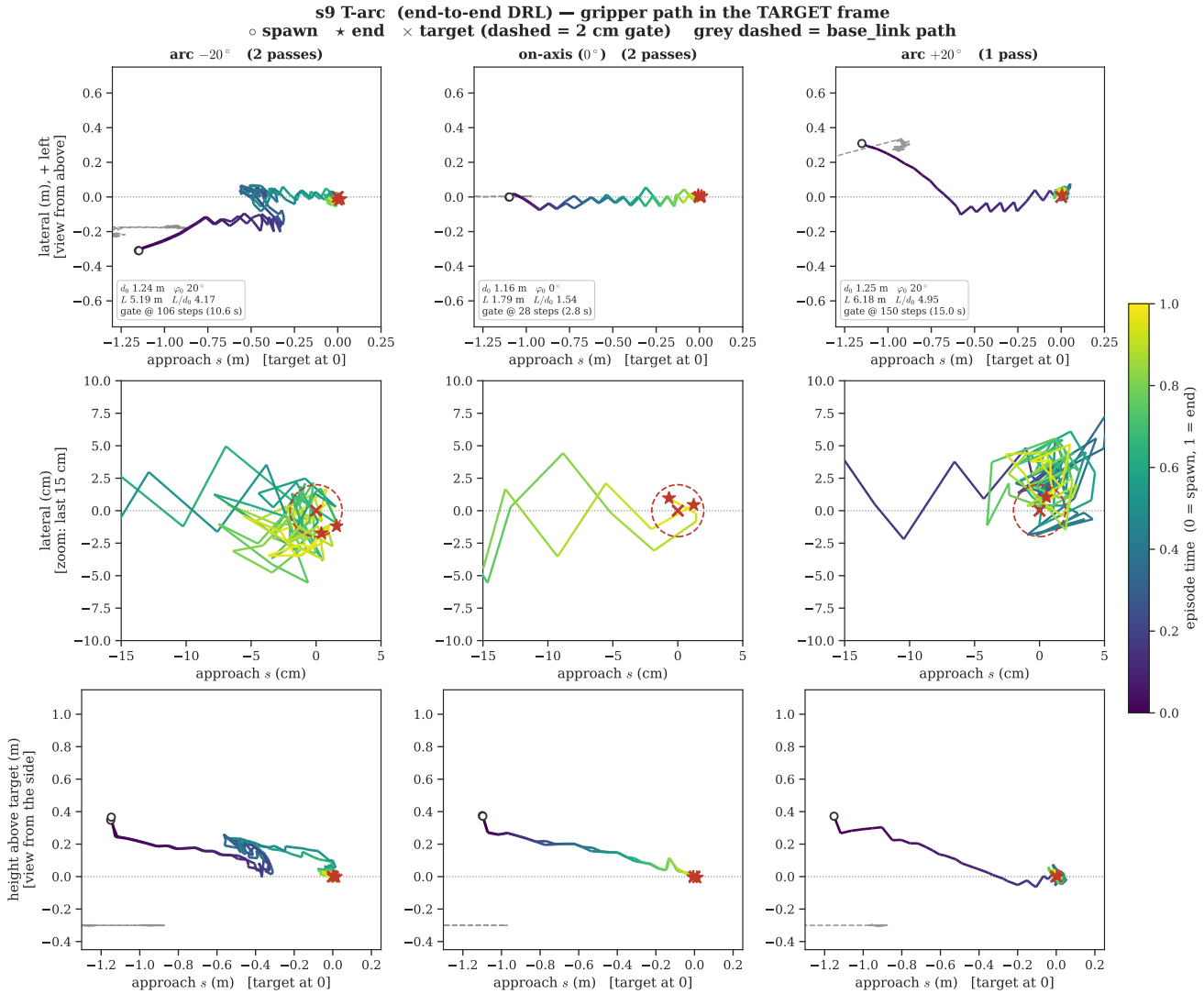

**What it shows.** Decisions to first enter the gate, per pass, annotated in seconds (1 decision = 0.1 s).

**How to read it.** Mean 83 decisions = 8.3 s. The on-axis target is reached in a fraction of the time the arc targets need.

**Say this.** "Mean 8.3 s to the gate over the sweep."

**If asked.** *Comparable to the WDLS regimes?* — yes: same three targets, same gate, same 0.1 s decision period. See `comparison_speed.pdf` at the top of `deliverables/`.

# Figure 6 — Trajectories in the target frame



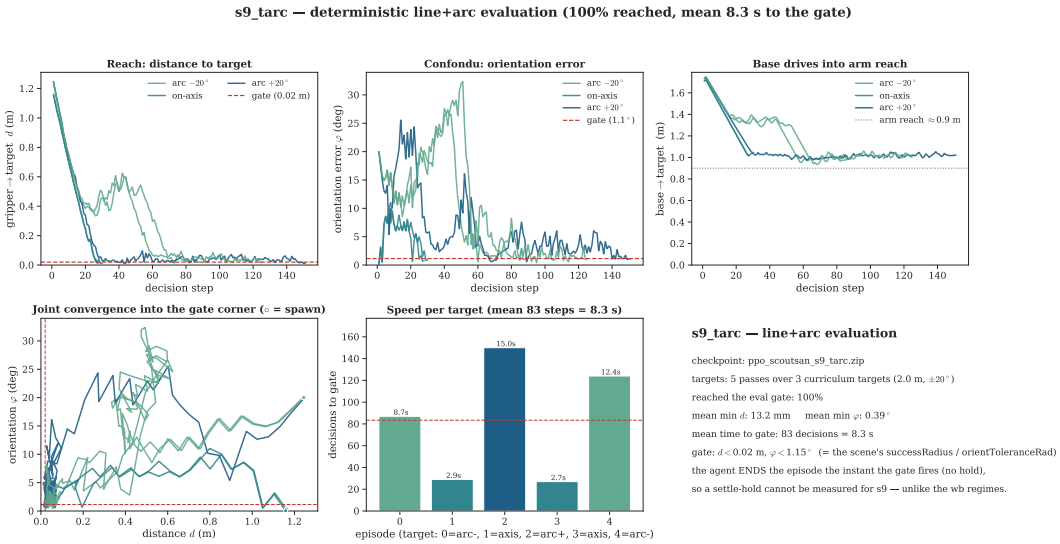

**What it shows.** The gripper's path expressed in the TARGET's own frame: $s$ = approach coordinate along the target's outward axis (the robot starts at $s < 0$, the target is $s = 0$), lateral = sideways offset, height = above the target. Colour = time within the episode. One column per curriculum target.

**How to read it.** Mean path length is 3.28× the straight-line distance, with up to 31 cm of lateral excursion on the arc targets; the zoom row shows the last 15 cm entering the 2 cm gate. The grey dashed line is where the BASE went — the base covers the long haul, the arm does the final alignment.

**Say this.** "Drawn in the target frame, the path is directly comparable with the other controllers — same targets, same axes, so the shapes can be read side by side."

**If asked.** *Why the target frame?* — the gripper→target relative pose is frame-independent, so it does not matter that the end-to-end model observes in base_link FLU while the WDLS logs are in Unity world coordinates. It is the only frame in which the two are comparable.

# Figure 7 — Overview (one-slide summary)




**What it shows.** Reach, orientation, base drive, phase portrait and per-target speed on one figure, plus the numbers.

**How to read it.** Use as the summary slide; the panels are detailed on the preceding pages.

**Say this.** "s9_tarc: 100% of the sweep reached, mean min $d$ 13.2 mm, mean min $\varphi$ 0.39°, 8.3 s to the gate."

**If asked.** *What does T-arc add over the fixed-target s9?* — the same agent and reward, trained on a curriculum that grows the target from close and on-axis out to 2 m and ±20°; this eval is the whole sweep, not a single pose.